



MULTIMODAL AI ENGINEER

Ibromhimjon Muminov

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M.S. in **Artificial Intelligence** with research focused on **Multimodal AI**, enabling AI systems to not only see the world but also reason about what they see to support more **natural human-machine interaction**.

Education

Dongguk University Master's degree in Artificial intelligence GPA: 3.85/4.5 Advisor: Prof. Jihie Kim	09/2024 – 08/2026 Seoul, South Korea
Moscow Technological Institute Bachelor's degree in Engineering GPA: 4.1/5.0 Advisor: Prof. Leonid Ivanovich Munkhoev	08/2016 – 06/2021 Moscow, Russia

Research Area

Multimodal AI: Vision-Language Models (VLMs), Visual Grounding, Multimodal Reasoning
Foundation Models: Large Language Models (LLMs), Language-Guided Object Detection
Computer Vision: Open-Vocabulary Object Detection, Computer Vision
Human-Centered AI: Human-Machine Interaction

Publications & Patents

Vag2Det: Handling Ambiguous Prompts in Knowledge-Based Open-World Detection, <i>Ibromhimjon Muminov, Jihie Kim</i> • (Accepted - AICompS) [2] <i>The International Conference on Artificial Intelligence Computing and Systems (AICompS 2025)</i> . 2025	11/2025
Commonsense-Guided Open-World ObjDet Using LLMs and Visual-Semantic Matching, <i>Ibromhimjon Muminov, Jihie Kim</i> • (Accepted - KISP) [1] Research and develop multimodal AI systems that leverage language reasoning to enhance open-world object detection, visual grounding, and human-centered scene understanding. <u><i>The Annual Symposium of KIPS 2025 (ASK 2025)</i></u> . 2025	05/2025
Semantic Planning and Candidate Reasoning for Visual Grounding, <i>Ibromhimjon Muminov, Jihie Kim</i> • (Under Review – ACCV 2026) Learning to Ground Like Humans: Semantic Planning and Candidate Reasoning for Visual Grounding. A cognitively inspired visual grounding framework (See & Think, Detect, and Decide) that models how humans resolve referring expressions through semantic planning and candidate comparison.	06/2026
Korean Patent Application • <i>Open-World Object Detection Using a Knowledge-Based LLM and Visual-Semantic Matching</i> . Filed with the Korean Intellectual Property Office (KIPO) as Korean Patent Application No. 10-2025-0070409 , filed May 29, 2025.	05/2025

Research Experience

Machine Learning Researcher, ITRC (대학정보통신연구센터협의회) - Conducting research in Multimodal AI, Large Language Models (LLMs), Vision-Language Models (VLMs), visual grounding, and open-vocabulary object detection. - Developing LLM-guided reasoning frameworks and multimodal datasets for visual grounding and language-guided object detection.	11/2024 – Present Seoul, South Korea
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Experience

Frontend Developer (AI Applications), iOyandasoz - Develop and maintain scalable web applications that integrate AI-powered services, ensuring reliable, efficient, and user-friendly experiences. - Integrate backend APIs for data processing and machine learning inference, building responsive interfaces to visualize model outputs and support AI-driven workflows.	02/2022 – 11/2022 Dushanbe, Tajikistan
Engineering Intern, Crocus Group - Worked with 2D/3D engineering drawings and CAD models, applying geometric reasoning, spatial analysis, and visual feature interpretation relevant to computer vision applications. - Processed and analyzed complex industrial layout models, leveraging 3D spatial understanding and geometric transformations applicable to computer vision tasks.	06/2017 – 12/2017 Moscow, Russia